Book 1 — Lecture 2 — Exercise 7

Orthogonality of Velocity and Acceleration in Uniform Circular Motion

Prompt. Show that the velocity and acceleration component patterns

$$v_x = -R\,\omega\sin(\omega t), \qquad v_y = R\,\omega\cos(\omega t),$$
$$a_x = -R\,\omega^2\cos(\omega t), \qquad a_y = -R\,\omega^2\sin(\omega t)$$

are orthogonal.[1]

Vectors

Write $\mathbf{v} = (v_x, v_y)$ and $\mathbf{a} = (a_x, a_y)$. Orthogonality means $\mathbf{v}\cdot\mathbf{a} = 0$.[2]

Dot product

Multiply termwise and add:

$$\begin{aligned}\mathbf{v}\cdot\mathbf{a} &= (-R\omega\sin(\omega t))\big(-R\omega^2\cos(\omega t)\big) + \big(R\omega\cos(\omega t)\big)\big(-R\omega^2\sin(\omega t)\big)\\ &= R^2\omega^3\sin(\omega t)\cos(\omega t) - R^2\omega^3\cos(\omega t)\sin(\omega t)\\ &= 0.\end{aligned}$$

Hence $\mathbf{v}\perp\mathbf{a}$ for all t (and all real ω, with the convention that the zero vector is orthogonal to every vector).

Related orthogonality (position and velocity)

For the same motion, $\mathbf{r}(t) = R\big(\cos(\omega t), \sin(\omega t)\big)$ has derivative $\mathbf{r}' = \mathbf{v}$. Moreover

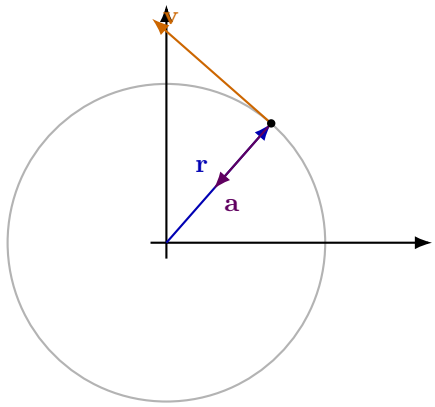
$$\mathbf{r}\cdot\mathbf{v} = R\cos(\omega t)\big(-R\omega\sin(\omega t)\big) + R\sin(\omega t)\big(R\omega\cos(\omega t)\big) = 0,$$

so *position* and *velocity* are orthogonal as well.

[1] These are the standard components of $\mathbf{v}$ and $\mathbf{a}$ for uniform circular motion with radius $R > 0$ and angular frequency ω (radians per unit time). Some editions of the notes misprint the heading as "position and velocity" instead of "velocity and acceleration"; the algebra below matches the latter. The word *acceleration* is sometimes mistyped in informal copies (e.g. "accelelration"). ω is spelled here (not "omege").

[2] Dot product and orthogonality are reviewed in Book 1, Lecture 1, Interlude 1, Exercise 6.

Picture. On the circle, $\mathbf{r}$ points outward along a radius, $\mathbf{v}$ is tangent, and $\mathbf{a} = -\omega^2\mathbf{r}$ points inward; thus $\mathbf{v} \perp \mathbf{r}$ geometrically, and $\mathbf{v} \perp \mathbf{a}$ because $\mathbf{a}$ is antiparallel to $\mathbf{r}$.



Result. With $\mathbf{v}$ and $\mathbf{a}$ as above, $\mathbf{v} \cdot \mathbf{a} = 0$, so velocity and acceleration are orthogonal. For $\mathbf{r} = R(\cos \omega t, \sin \omega t)$, also $\mathbf{r} \cdot \mathbf{v} = 0$.

Example. At $\omega t = \pi/4$: $\mathbf{v} \propto (-1, 1)$, $\mathbf{a} \propto (-1, -1)$; their dot product is zero.